

Single Sample t Test

ITAO 20200: Foundations of Statistics

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Description

A single or one-sample t test examines how the mean of an individual sample compares against a known population mean or value. For instance, suppose there is a new format for taking the SAT. We might be interested in determining if the average score for a sample who took the test under the new format is different from the population mean score obtained from all people who took the SAT with the old exam format.

Assumptions

The variable under investigation must be measured on either an interval or ratio scale and should be normally distributed. However, the t test is robust to minor violations of the assumption of normality.

Data

For this test, just one variable is required, so only a single column of data is necessary. In this example, we will use the **Grades.csv** data. There are five variables in this data set. **Pretest**, **Midterm**, and **Final** all represent scores obtained on those tests. **Teacher** is a grouping variable indicating which of three different instructors taught the students. **Required** is another grouping variable indicating whether or not the class was an elective or was required. A portion of the data is shown below.

```
Grades <- read.csv("https://mendoza-stats.s3.amazonaws.com/grades.csv")
head(Grades, n = 5) # First 5 observations in the dataset
```

##	Pretest	Midterm	Final	Teacher	Required
## 1	56	75	77	1	1
## 2	79	91	84	1	0
## 3	68	82	81	1	1
## 4	57	72	72	1	0
## 5	70	81	76	1	1

Data Analysis & Output

Our objective is to determine if scores on the pretest differ from expected population values.

Some Summary Statistics

First, let's obtain some basic summary statistics about our variable of interest. The `summary` command provides several descriptive statistics: minimum, maximum, median, mean, and the first and third quartiles. The `sd` command returns the sample standard deviation. The `length` command indicates the sample size for that variable.

```
summary(Grades$Pretest)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   50.00   57.00   61.00   64.12   70.25   80.00
```

```
sd(Grades$Pretest)
```

```
## [1] 8.573531
```

```
length(Grades$Pretest)
```

```
## [1] 24
```

One Sample t Test

Now, let's move on to do our t test. R is capable of doing either one- or two-tailed t tests using the `t.test()` function. The argument that differentiates between the type of test is `alternative`. For a two-tailed test, use `alternative = "two.sided"` (which is the default setting). For a lower-tailed hypothesis test, use `alternative = "less"` option in the function. For an upper-tailed hypothesis test, use `alternative = "greater"`.

We would like to test the following hypotheses at $\alpha = 0.05$:

$$H_0 : \mu = 70$$

$$H_a : \mu \neq 70$$

To do this in R, use the `t.test()` function as follows:

```
t.test(Grades$Pretest,           # the variable we are testing
       alternative = "two.sided", # the alternative hypothesis
       mu = 70,                  # the null value
       conf.level = 0.95)        # the confidence level = 1 - alpha
```

```
##
## One Sample t-test
##
## data:  Grades$Pretest
## t = -3.357, df = 23, p-value = 0.002728
## alternative hypothesis: true mean is not equal to 70
## 95 percent confidence interval:
##  60.50472 67.74528
## sample estimates:
```

```
## mean of x
##      64.125
```

The output is labeled **One-Sample t-test**. In the result:

- **t** is the obtained test statistic value
- **df** is the degrees of freedom associated with the t test
- **p-value** is the obtained p -value for the t test
- **alternative hypothesis** specifies the form of hypothesis we are testing
- **95 percent confidence interval** provides the 95% CI of the true mean value
- **sample estimates** reports the mean of the sample

Interpreting the Output

If the reported p -value is less than or equal to your alpha level ($p \leq \alpha$), then there is a statistically significant difference between the sample mean and the null value. There is a very low probability of obtaining such an extreme value for t if, in fact, the null hypothesis is true. If the reported p -value is greater than your alpha level ($p > \alpha$), then your results are inconclusive, and no significant difference has been found.

The confidence interval can also be used to determine if a statistically significant result exists. Evaluate the interval with respect to the null value (in this case, 70). Recall that the interval represents a range of plausible mean values for your sample at the population level. If the null value falls outside of the interval, then it is unlikely to represent this sample at the population level. You can conclude that there is a significant difference between your sample and the null value. If the null value falls inside of the interval, then it could possibly represent this sample at the population level. Therefore, a null value that falls inside the boundaries of a confidence interval indicates no significant difference on your t test.

Writing Your Results

To convey your results properly, you need to know the t value, the degrees of freedom, and the p -value of the test statistic. Additionally, it would be helpful to have the mean and standard deviation of the sample in question. In the example above, we could report a statistically significant result.

A single sample t test was performed to determine if average score on the Pretest differed from the expected value of 70. Results indicate that the average score on the Pretest ($\bar{x} = 64.12$, $s = 8.57$) was significantly lower than expected, $t(23) = -3.36$, $p = 0.0027$.

Practice On Your Own

1. Obtain some summary statistics about **Midterm Scores** and determine if the average score is *less than* a population value of 84 using $\alpha = 0.05$, i.e

$$H_0 : \mu \geq 84$$

$$H_a : \mu < 84$$

2. Obtain some summary statistics about **Final Scores** and determine if the average score is *greater than* an expected value of 85 using $\alpha = 0.05$, i.e

$$H_0 : \mu \leq 85$$

$$H_a : \mu > 85$$